TELEDRIVE – STEP-BY-STEP WORKING SUMMARY

TeleDrive is a smartphone-based edge telematics system designed for Indian two-wheelers, especially gig-economy riders. The system uses only the sensors already available in a standard Android smartphone to analyze riding behavior, safety, and fuel efficiency without installing any external hardware.

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1. System Start and Ride Detection

TeleDrive runs silently in the background.
When the rider starts moving and the GPS speed crosses a minimum threshold, the system automatically detects that a trip has begun. No manual interaction is required from the user.

This design ensures the app does not distract the rider during driving.

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2. Sensor Data Collection

During the ride, the smartphone continuously collects data from:

- Accelerometer
- Gyroscope
- GPS

These sensors capture motion, orientation, speed, and acceleration patterns of the vehicle.

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3. Virtual Reorientation of the Smartphone

Since riders may keep their phone in different positions (handle mount or pocket), raw sensor data cannot be directly trusted.

To solve this, TeleDrive applies a virtual reorientation algorithm that aligns the phone's sensor axes with the vehicle's movement direction.
This allows consistent interpretation of riding behavior regardless of phone placement.

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4. Signal Processing and Noise Filtering

Indian two-wheelers generate strong engine vibrations that introduce noise into sensor readings.

TeleDrive applies digital low-pass filtering to remove high-frequency vibration noise while retaining meaningful riding signals such as:

- Acceleration
- Braking
- Swerving

This step improves accuracy and reduces false detections.

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5. Edge AI Riding Event Detection

The filtered sensor data is analyzed using a lightweight 1D Convolutional Neural Network (1D-CNN) deployed on the smartphone using TensorFlow Lite.

The model classifies riding patterns into:

- Safe riding
- Rash riding (harsh braking, aggressive weaving)
- Road irregularities (potholes, speed breakers)

All AI processing happens locally on the device, ensuring low latency and strong privacy.

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6. Eco-Efficiency Analysis using Vehicle Specific Power (VSP)

TeleDrive estimates fuel efficiency using Vehicle Specific Power (VSP), a physics-based metric that represents how much power the vehicle is demanding per unit mass at any moment.

VSP is calculated using:

- Vehicle speed
- Acceleration

High VSP values indicate aggressive or inefficient riding, which correlates with higher fuel usage.
Low VSP values indicate smooth, fuel-efficient riding.

The system does not measure exact fuel consumption. Instead, it estimates fuel waste trends based on riding behavior.

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7. Eco-Score and Fuel Waste Estimation

Cumulative high-VSP periods are converted into an Eco-Score that reflects riding efficiency.

An estimated fuel waste value may be shown using average vehicle efficiency assumptions. This value is approximate and meant for awareness and comparison, not precise measurement.

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8. Post-Trip Summary and Visualization

After the ride ends, TeleDrive presents a trip summary that includes:

- Safety score
- Eco-Score
- Visual indication of inefficient riding zones

No alerts or feedback are shown during riding, ensuring safety and zero distraction.

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9. Privacy-First Data Handling

TeleDrive follows a privacy-first, edge-computing architecture.

Raw sensor data and continuous location trails remain on the device.
Only summarized scores are stored or shared, making the system compliant with modern data protection principles.

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10. User Benefit and Practical Use

For gig workers, fuel cost and safety directly affect daily income.
TeleDrive helps riders identify inefficient habits and improve riding behavior without surveillance, hardware costs, or employer pressure.

The system encourages safer and more economical riding through insight and optional gamification rather than punishment.

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11. Overall System Conclusion

TeleDrive demonstrates that a standard smartphone can function as a secure, low-cost, edge-AI telematics unit for Indian two-wheelers.
It provides meaningful safety and fuel-efficiency insights while respecting privacy and real-world constraints.